Avery Briggs | abriggs914@gmail.com

Professional Summary

Software developer with a Bachelor of Computer Science and experience building internal software systems, automation platforms, and operational dashboards using Python, SQL, and modern data processing libraries. Experienced in developing full-stack applications, data pipelines, and integration tools used across multiple departments within a manufacturing environment. Strong background in relational databases, systems design, and building practical software that supports real-world operational workflows. Actively expanding expertise in modern web development technologies including TypeScript and .NET.

Experience

Software Systems Developer | Process Analyst

BWS Manufacturing Ltd. | Centreville, N.B. (Jan 2021 – Feb 2026)

Developed internal software systems and software development tools supporting operational workflows across Sales, Procurement, Production, and Scheduling within a small technical team in a manufacturing environment.

Key Contributions

- Full-stack development of Python-based internal applications and Streamlit dashboards used by multiple departments to analyze operational data and automate workflows and support decision-making.
- Developed complex SQL Server data pipelines and analytics queries enabling cross-department analysis of inventory, production, purchasing, and sales metrics.
- Built automation tools that eliminated manual operational tasks, including systems that automatically aggregate and prepare files for reporting and meetings, reducing manual preparation time from hours to seconds.
- Designed and implemented a warehouse visualization system mapping inventory locations onto spatial layouts to improve operational visibility and planning.
- Developed tools using OCR techniques to extract structured data from legacy PDF documents, enabling automated reporting and analysis of previously inaccessible operational records.
- Refactored legacy Microsoft Access applications by consolidating duplicated VBA logic into reusable shared modules, eliminating 3–5 smaller databases and improving maintainability of the primary operational system.
- Built an administrative time-tracking interface integrating live ShopClock data, allowing supervisors to review, adjust, and approve employee shift records before payroll submission, improving visibility and coordination across multiple departments.
- Created and maintained reusable Python utility libraries supporting database access, file management, reporting, and UI helpers used across internal projects.
- Provided technical troubleshooting and support across hardware systems, network devices, ERP integrations, and vendor-managed services.

Notable Projects:

- IT Request and Inventory Ticketing System. Built from scratch using SQL and Access.
- Custom drag-n-drop production schedule. CustomTkinter application packaged into an executable file.
- Streamlit dashboards for sales margins and inventory analysis including inputs and reporting.
- 3D warehouse visualization using Plotly, demonstrating inventory bottlenecks and improving transparency.
- PDF parsing of legacy reports. Performed using Python and modules: Pdfplumber, Streamlit, and Pytesseract.
- Implementation of Access version control protocols to maintain program integrity.
- Yellow Tag system to automatically flag, record, and report missed parts when picked for production.

Technologies used:

- Python, SQL Server, Streamlit, Pandas, Plotly, Pytesseract, Reportlab, CustomTkinter, Microsoft Access, VBA, Syspro8 ERP, ShopClock, NeWare Security

Other Job Experience:

- Basketball Referee – CBOC – Fredericton, NB – Oct 2013 – Mar 2020
- Kitchen Staff – FMI – Fredericton, NB, -- May 2015 – Sept 2017
- Farm Hand – Hunter Brother’s Farm Market – Florenceville, NB, -- Jun 2011 – Aug 2013

Avery Briggs | abriggs914@gmail.com

Education

Bachelor of Computer Science

UNB – Fredericton N.B. (May 2020)

- Specialization: Software Systems.
- Recipient of Governor Thomas Carleton Entry Scholarship.

Skills

- **Programming**: Python, SQL, VBA, Java
- **Frameworks / Libraries**: Streamlit, Pandas, Plotly, Tkinter, CustomTkinter, Pdfplumber, Pytesseract
- **Learning**: TypeScript, C#, .NET, other modern web frameworks
- **Databases**: Microsoft SQL Server, Microsoft Access
- **APIs**: NHL Stats API, IMDB API, Fitbit API
- **Tools:** Git, Syspro ERP, ShopClock
- **Languages**: English (native), French (Intermediate oral proficiency)
- **Other**: Strong cross-team communication and collaboration. Self-directed problem solving and rapid learning.

Awards & Activities

- Finalist (Top 12) - Kaspersky Academy Partner's Summit (KAPS) Lisbon, Portugal (December 2019).
- Certified intermediate plus in the French oral proficiency exam (June 2013).
- UNB dormitory Orientation Committee (September 2014) & Social Representative (September 2014 – April 2015).
- UNB Relay for Life steering committee logistics team (September 2013 – February 2014).

Hobbies

3D printing and hardware projects, sports analytics software, recreational athletics (cycling, kayaking, basketball officiating)

Projects

- Live NHL Scoreboard showcasing game scores and standings data, with an interactive playoff bracket selection. Leverages NHL Stats API to show real-time figures, summarized
- STL file generation for automating the creation of strings of text using custom fonts and other parameters. Uses a Solid2 to generate files, and Streamlit interface to model and tweak them.
- Minesweeper Android mobile application, built using Java and Android Studio.
- Image editing Android application taking advantage of the Gestures library to offer a unique interface for tuning photos.
- Traffic simulation engine using Pygame. Mimics some elements from Cities Skylines traffic patterns.
- COVID-19 interaction simulator built with JavaFX modeling contagion spread in configurable populations.

References available upon request